

Run report: every scenario, every arm

The recorded run, scenario by scenario: the naive arm, the controlled arm, and what each one cost. Simulation units.



01 + 02 · Production AI Engineering

4 min read · Published 2026-10-08 · Run report

The run

Ten scenarios and a negative control, recorded once from the frozen preregistration (seed 4917), replayed byte for byte. Every row below is aggregated from the run's raw telemetry rows
(ops_poc/runs/2026-10-08-recorded/scenarios/).

E1 · Admission (Black Friday surge)

	open admission	bounded admission
support requests	2,640	2,640
attempts, retries included	7,430	4,564
attempts per request	2.8	1.7
max work in system	5,791	128
max queued	5,759	96
queue delay p50 (sim-ms)	365,808	11,589
queue delay p99 (sim-ms)	686,675	14,027
latency p95, successes (sim-ms)	683,415	23,374
refused explicitly	0	1,405
served while the client waited	245	1,200
goodput (%)	9.3	45.5
cost on attempts nobody waited for (%)	97.4	57.9
total cost (cu)	111,168.4	27,966.3

E2 · Tenant fairness (support + a finance batch)

	one global FIFO	fair scheduling + bulkhead
support served in time (%)	3.6	100
support queue delay p99 (sim-ms)	351,027	499
support attempts per request	2.9	1
finance items completed	3,000	3,000
finance makespan (sim-s)	168.1	374.9
finance running at once, max	32	16
support running at once, max	32	29

E3 · Bounded concurrency and deadlines

	unbounded	slots + bounded queue + deadlines
running at once, max	1,227	32
goodput (%)	44.2	84.4
latency p95, successes (sim-ms)	251,507	16,388
cost on attempts nobody waited for (%)	95.2	7.8
dropped at dequeue (deadline)	0	36
cancelled at the deadline	0	1
refused explicitly	0	132
started after the client's deadline	0	0

E4 · Workflow resource envelopes

	no budget	per-agent budgets	propagated envelope
runaway refund: result	HARNESS_GUARD	BUDGET_EXCEEDED	BUDGET_EXCEEDED
runaway refund: steps	400	12	12
runaway refund: cost (cu)	18,960.3	150	150
coordinator + 3 sub-agents: result	HARNESS_GUARD	BUDGET_EXCEEDED	BUDGET_EXCEEDED
coordinator: model calls, children included	205	11	9
coordinator: cost (cu)	16,735.1	153.1	127.6
coordinator: stopped on	none	wall_ms	fanout
legitimate workflows cut	0	4	4

Cut by the envelope: delivery-change (steps); dispute-investigation (wall_ms) (4 of them had re-run after a wrong answer). The envelope (refund dispute): 18 steps, 8 model calls, 5 tool calls, 184.7 cu; coordinator: 10 model calls.

E5 · Routing inside an eligibility contract

	all-large	all-small	routed	routed, any fallback
success (%)	96.4	98	99.1	100
first answer wrong (caught)	30	81	33	33
escalated	0	16	0	0
deferred during the throttle	29	0	7	0
cost per request (cu)	24	2.5	9.9	10.3
cost per success (cu)	24.9	2.6	10	10.3
... with escalations at 60 cu	24.9	3.8	10	10.3
calls below the capability tier	0	516	0	0
calls breaking the data policy	0	0	0	18
fallbacks	0	0	0	18
latency p95 (sim-ms)	11,404	4,691	11,404	11,404

Break-even: all-small stays cheaper per success than routed unless one escalation costs more than 364 cu (under this declared contract and success table).

E6 · Context budgets and cache scope

	naive	bounded
context tokens per query (mean)	34,716	5,982
chunks retrieved (12 queries)	960	96
source queries	48	24
required evidence ids missing	0	0
retrieval cost (cu)	2.9	0.7

cache	lookups	hits	served across principals	served stale
keyed by query text	24	15	5	5
scoped key	24	7	0	0

E7 · The tool gateway

	direct calls	gateway	gateway + fair scheduling
payments calls in flight, max	16	8	8
503s	237	0	0
429s	5,932	0	0
attempts per logical call	2.8	1	1
workflows completed	911	3,584	2,222
workflows failed on the tool	1,311	0	0
support served in time (%)	99.2	5.3	100
finance items completed	195	1,500	1,500
finance makespan (sim-s)	31.8	157.6	161.2

E8 · The behavioural release (same code, seven releases)

release	changed artifact	release id	tool calls/wf	input tokens/wf	cost/wf (cu)
R41	(production)	rel-c084dffaf6c9	1.1	8,580	15.1
R42-e	prompt	rel-0e00bddbbfffb	1.1	7,789	14
R42-a	tools.orders.lookup	rel-ec085ba7c34e	1.9	8,876	15.2
R42-topk	retrieval	rel-550efbb7626c	1.1	12,020	20.5
R42-kb	knowledge	rel-3b3cce63002f	1.1	8,580	15.1
R42-policy	authz_policy	rel-91e7d99815c9	1.1	8,580	15.1
R42-routing	routing	rel-d540123fc17c	1.1	8,562	18.5

Image digest, all seven: img-b07d496787c0. Telemetry rows with a release id: 58,573 of 58,573.

E9 · The offline gate (40 cases)

candidate	task success	data policy	approval bypass	envelope	cost Δ (%)	gate
R42-a	1	0	0	0	0.7	none
R42-b	1	21	0	0	-15.7	invariant data_policy: 21
R42-c	1	0	4	0	0	invariant approval_bypass: 4
R42-d	1	0	0	1	0	invariant envelope_unenforceable: 1
R42-e	1	0	0	0	-7.2	none

E10 · The canary

analysis	decision	windows	at (sim-s)	last window: tool calls Δ (%)	cost/success Δ (%)	error Δ (pp)	breaches
R42-a.mix-adjusted	ROLLBACK	1	132.7	88.9	-0.1	0	tool_calls_pct
R42-e.mix-adjusted	PROMOTE	4	320.6	4.4	-5.9	0	none
R42-e.raw	ROLLBACK	3	297.8	8.8	21	0	cost_per_success_pct

R42-a: replies sent before the rollback 60, undone 0; requests to R42-a after the decision 0.

Negative control

E1's controlled arm without the admission bound: max work in system 5,791 against a bound of 128; the check OPS-NC-C01 failed, as it must.

Raw files: `ops_poc/runs/2026-10-08-recorded/` · every recorded decision explained: `uv run agentops explain E10-canary` (and E1-surge, E4-runaway, E5-routing, E8-release, E9-gate).

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The full learning map (Start Here)

O1 + O2 · Production AI Engineering

Production AI Engineering · O1 + O2. Every measured number here is substituted at build time from ops_poc/runs/2026-10-08-recorded/facts.json. The platform controls are real code; the clock, providers, tools, clients and agents are a deterministic simulation in simulated milliseconds and cost units, and the whole run replays byte for byte. Figures carry their provenance.

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